House Price Prediction Using Machine Learning

Project Overview:

The objective of this project was to build a machine learning model capable of predicting house prices based on various property features. The dataset was loaded using Pandas and explored to understand its structure and characteristics. The target variable was identified as **price**, while the remaining columns were used as features for prediction.

Data Cleaning and Preprocessing:

The dataset was examined for missing values and duplicate records. No missing values or duplicate rows were found, so no imputation or row removal was required. Several columns contained categorical values such as main road, guestroom, basement, air-conditioning, and furnishing status. These columns were converted into numerical format using one-hot encoding so that they could be used by machine learning algorithms.

Model Building:

The dataset was split into training and testing sets using an 80:20 ratio. Two machine learning models were trained and evaluated:

1. Linear Regression
2. Random Forest Regressor

Model performance was measured using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score.

Results:

Model	MAE	RMSE	R^2 Score
Linear Regression	970,043	1,324,507	0.653
Random Forest Regressor	1,021,546	1,400,566	0.612

The Linear Regression model achieved better performance across all evaluation metrics. It produced lower prediction errors and a higher R^2 score than the Random Forest model.

Key Findings:

The correlation analysis indicated that features such as area, bathrooms, stories, parking availability, and furnishing status had a significant influence on house prices. The Linear

Regression model explained approximately 65.3% of the variation in house prices, demonstrating a reasonable predictive capability for this dataset.

One interesting observation was that the simpler Linear Regression model outperformed the more complex Random Forest model. This suggests that the relationship between the selected housing features and price is largely linear.

Recommendation:

Based on the analysis, real estate businesses should pay particular attention to factors such as property area, number of bathrooms, number of stories, parking availability, and furnishing quality when estimating house prices. These features appear to have the strongest impact on property value and can be used to support more accurate pricing decisions.